

Hrithik Puri

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EXPERIENCE

Serigor

Mar 2025 - Present

Software Engineer

Baltimore, Maryland

- Architected a **GenAI** Tax model using **Python**, **LangChain**, **RAGAS**, **Weaviate** on Financial Data, setting up guardrails ensuring compliance with tax rules, boosting tax validation accuracy and reducing manual audit review time by 30%
- Built an AI prototype with **Python**, **React**, and **Hugging Face Transformers** for text analysis, deploying model through **FastAPI**-based REST services to support scalable text analysis workflows and improving inference throughput
- Optimized stored procedures on **SQL Server Management Studio** to reduce time for matching MEDRED/MEDFREE students by 70% and created dashboards on **Power BI Report Server** analyzing data for 600K+ enrolled students

Pinnacle IT Solutions

Nov 2024 - Feb 2025

Data Scientist

Brighton, Colorado

- Designed a ML based patient churn classification model using **Python**, **Scikit-learn**, leveraging **Decision Tree** on healthcare data to identify high-risk patients, reducing patient attrition by 35% and improved retention revenue
- Implemented **MLOps** practices by integrating CI/CD-driven **Docker** deployments, automating model packaging and versioning, resulting in faster release cycles and improved consistency
- Developed and automated **ETL pipelines** with **Python**, **Airflow (DAGs)** and **Snowflake** for healthcare data integration, cutting processing time by 60% and supporting compliance with HIPAA standards

Veeco Instruments

Jul 2023 - Dec 2023

Data Scientist

San Jose, California

- Developed a **Multimodal CNN-RNN** neural network architecture for wafer breakage detection on Laser Spike Annealing (LSA) tool using **PyTorch**, **Keras**, and **Scikit-learn**, improving inline defect detection accuracy by 25% and reducing yield loss
- Orchestrated ETL pipelines to set up **Data Infrastructure** for process tools to ingest high-frequency fab, metrology, and sensor data, reducing data latency by 45% and enabling real-time semiconductor process analytics
- Conducted **statistical analysis** using **Pandas**, **NumPy** on semiconductor **manufacturing data** to detect lithography and laser performance trends during fabrication, resulting in a 20% increase in overall throughput efficiency

SYMB Technologies

Sep 2021 - Feb 2022

Software Engineer

Noida, Uttar Pradesh

- Engineered query based data pipelines with **Python**, **Flask**, **PostgreSQL**, improving **REST APIs** response time by 30% and enabling real-time analytics that accelerated reporting cycles
- Deployed **CI/CD workflow** using GitHub Actions on **AWS resources (S3, ECR, EC2)**, automating build, test, and deployment processes, reducing deployment time by 40%

Ducat India

Apr 2021 - Aug 2021

Data Analyst

Noida, Uttar Pradesh

- Created **Tableau** dashboards for 300K student feedback records, that improved program effectiveness by 20%
- Conducted **A/B testing** with over 5,000 participants, identifying a 22% improvement in student engagement

SKILLS AND CERTIFICATIONS

- Programming Languages:** Python, R, SQL, C/C++, React, JavaScript
- Analytics & Visualization:** Matplotlib, Pandas, NumPy, Excel, Tableau, Power BI, A/B Testing, Statistics
- AI & ML Frameworks:** Scikit-learn, TensorFlow, PyTorch, LangGraph, LangChain, Neural Networks, Transformers, FastAPI
- Data Engineering:** ETL Pipelines, Apache Airflow, PySpark, dbt, Databricks, DevOps, CI/CD, MLOps, LLMops
- Databases:** MySQL, Snowflake, PostgreSQL, MongoDB, Pinecone, Weaviate, Cassandra
- Tools:** Git/GitHub, Notebooks, Kubernetes, Docker, Terraform, AWS, GCP, Azure

PROJECTS

Enterprise Knowledge Assistant | [Github Link](#)

- Built and deployed a RAG-powered agent using **Python**, **Gemini API**, **LangChain**, **FAISS**, **Streamlit**, **Docker**, reducing employee document search time by 65%, accelerating decision-making across enterprise users

Derma Vision | [Github Link](#)

- Developed a CNN-based classification model using **ResNet50** and custom CNN architecture for skin lesion classification, achieving 67.76% test accuracy by data augmentation, cross-entropy loss optimization, and L2 weight decay regularization

EDUCATION

Northeastern University

Sep 2022 - Aug 2024

Master of Science, Data Analytics Engineering

Uttar Pradesh Technical University (UPTU)

Aug 2018 - Jul 2022

Bachelor of Technology, Computer Science Engineering

CERTIFICATION

- AWS Academy Graduate - Cloud Architecting, Cloud Foundations